

Operating Systems

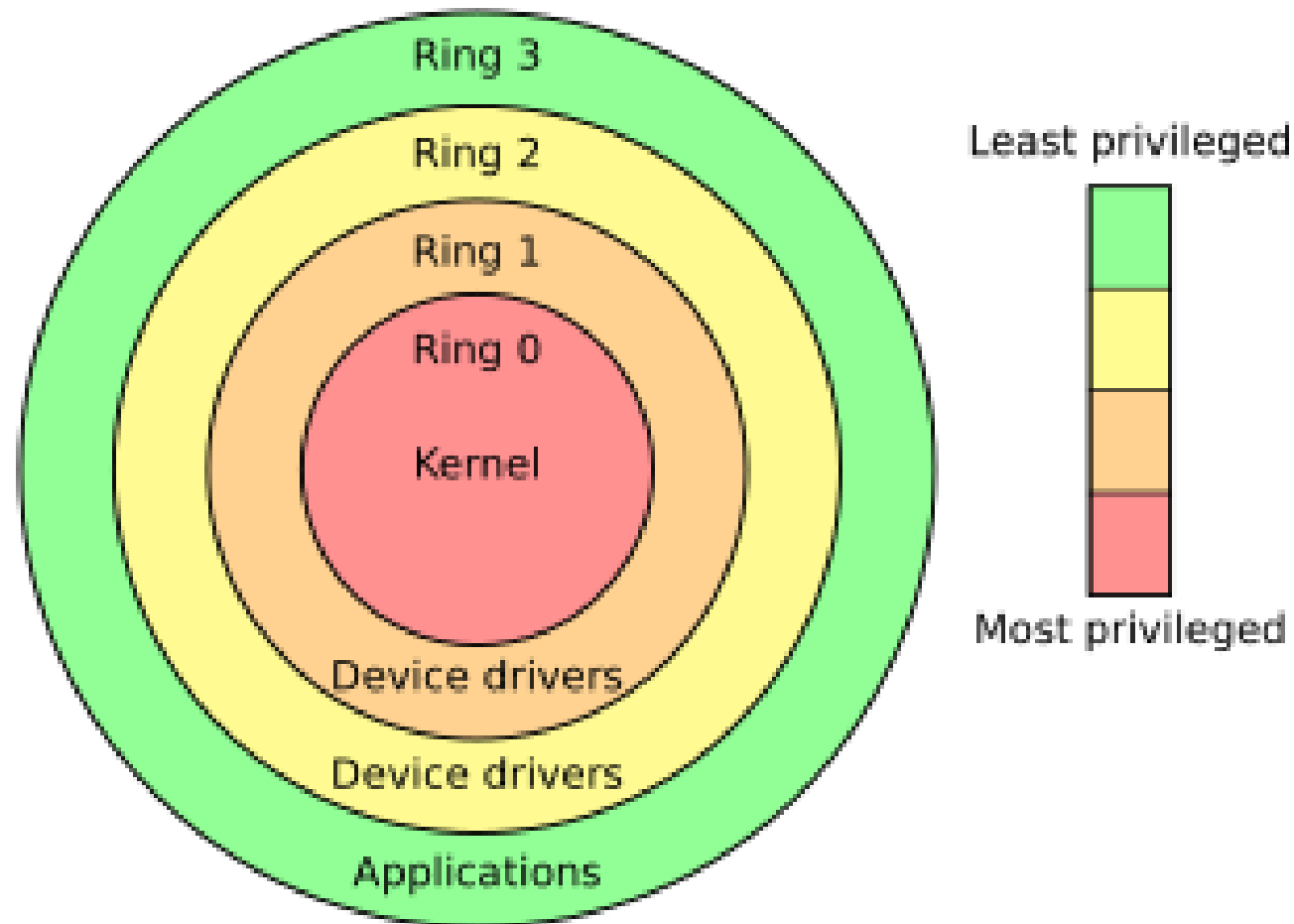
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Lecture 23: Information protection in OS. Protection levels and context switching.

Privilege levels of CPU

- Current Privilege Level (CPL) of the running code (Ring 0–3)
- IOPL (I/O Privilege Level) is a 2-bit field in the RFLAGS register (bits 12 and 13) that controls whether a program can directly communicate with hardware peripherals (IN and OUT instructions).
- Access Granted: If $CPL \leq IOPL$, the code can execute I/O instructions.
- Access Denied: If $CPL > IOPL$, the processor triggers a General Protection Fault (#GP)

Protection ring

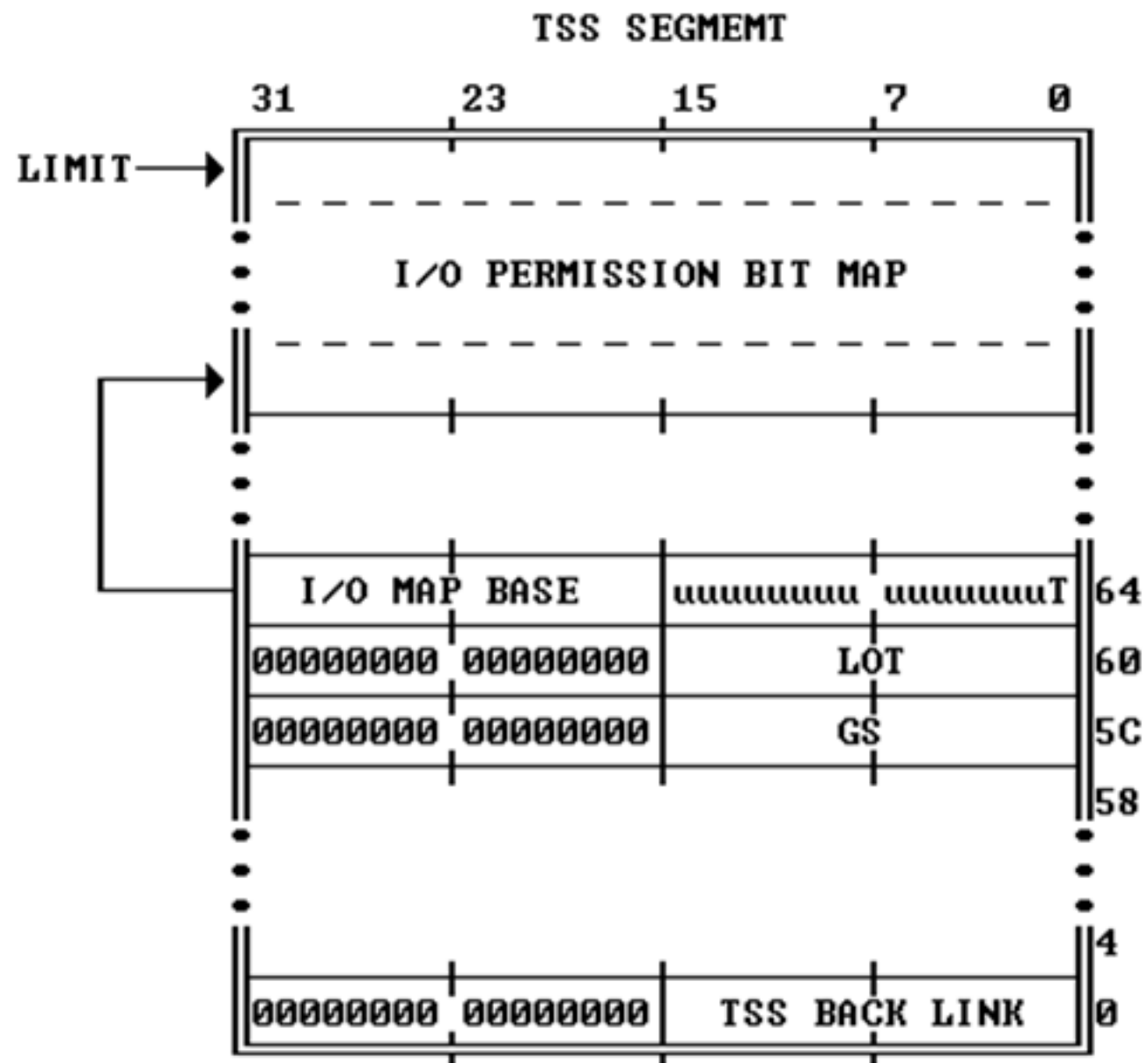


How to change privilege levels

- The IOPL can only be changed by code running at Ring 0. You cannot elevate your own I/O privileges from a user-level program; you must ask the OS kernel (e.g., using the `iopl()` system call in Linux) to change it for you
- In modern x86_64 systems, there is a second layer of protection called the I/O Permission Bitmap located in the Task State Segment (TSS). Even if your $CPL > IOPL$, the OS can use this bitmap to grant you access to *specific* hardware ports while keeping others locked

Permission bit map

I/O Address Bit Map



The Granular Check (I/O Permission Bitmap)

- If you fail the global check (which is common for Ring 3 applications), the CPU looks at the Task State Segment (TSS).
- Inside the TSS is the I/O Permission Bitmap.
- Each bit in this map represents one of the 65,536 possible I/O ports.
- 0: Access is allowed to that specific port.
- 1: Access is denied, and a General Protection Fault is triggered.

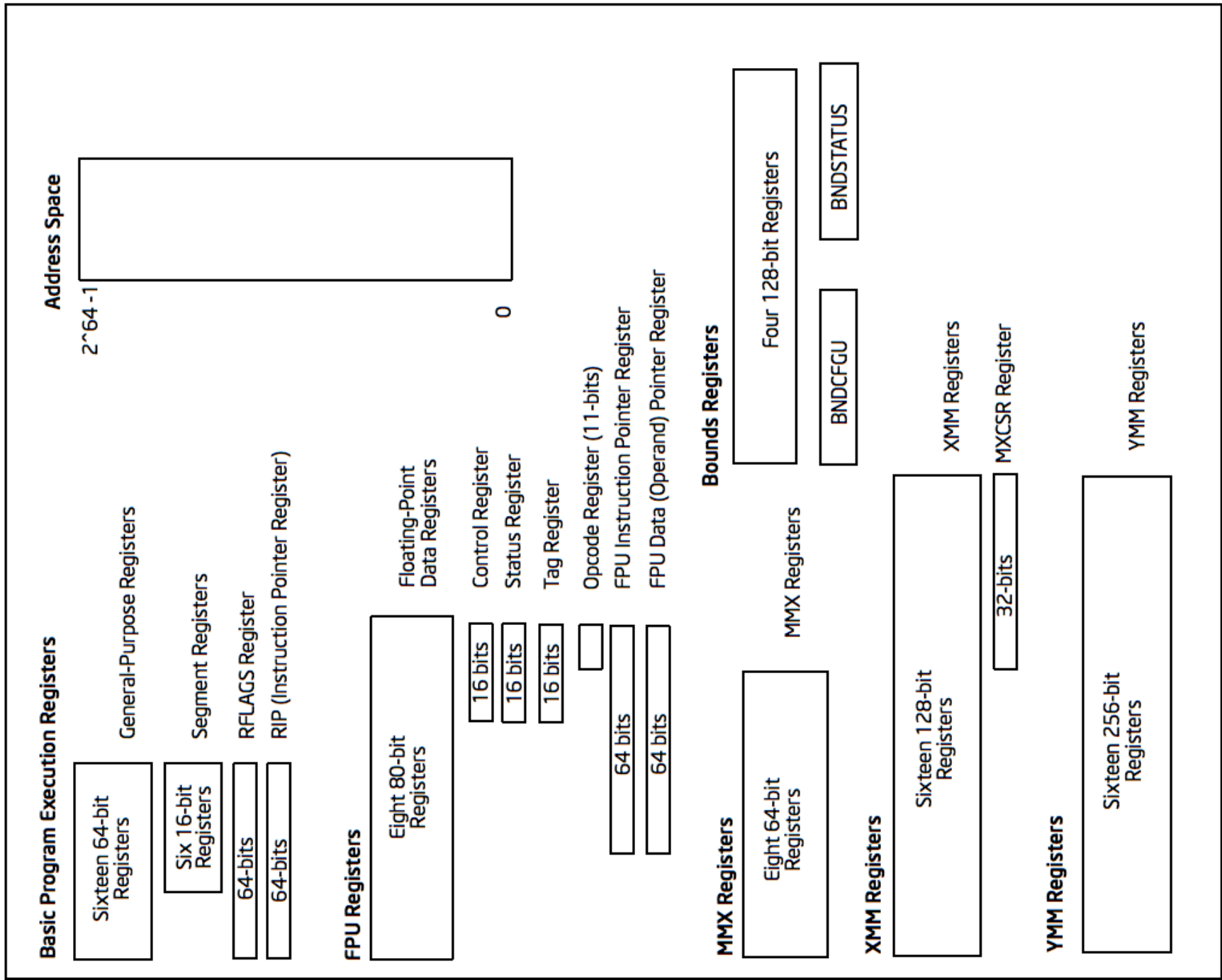
Control Registers (CR0–CR8)

- CR0: Controls basic operation (e.g., switching on Protected Mode, enabling the Cache).
- CR2: Stores the Page Fault Linear Address (the memory address that caused a crash).
- CR3: Holds the physical address of the Page Directory (the root of the virtual memory map).
- CR4: Enables advanced features like Virtual Machine Extensions (VMX) or large pages.
- Access: Only accessible at Ring 0 (Kernel) using MOV instructions (e.g., MOV CR0, RAX).

Descriptor Table Registers

- GDTR: Global Descriptor Table Register.
- IDTR: Interrupt Descriptor Table Register (points to the Interrupt Vector Table).
- LDTR / TR: Local Descriptor Table and Task Register.
- Access: Loaded using LGDT / LIDT and read using SGDT / SIDT, Ring 0.

The basic architecture of the x86-64



Memory protection – Page protection bits

- Every 4KB page has a set of attribute bits that define what is allowed:
- R/W (Read/Write): If 0, the page is Read-Only. If 1, writing is allowed.
- U/S (User/Supervisor): If 0, only the Kernel can access it. If 1, User apps can access it.
- NX (No-Execute): The 63rd bit. If set, the CPU will refuse to run code from this page (prevents buffer overflow exploits).
- P (Present): If 0, the page isn't in RAM. Accessing it triggers a Page Fault.

What happens on memory access violation

- If a program tries to write to a read-only page or access Kernel memory:
- The CPU stops the instruction immediately.
- It saves the failing address in the CR2 register.
- It triggers an Interrupt 14 (#PF - Page Fault).
- The OS Kernel takes over to either load the page or kill the crashing program (Segmentation Fault).

Protection of info on hard drive

- Encryption, Access Control, and Hardware Locking
- Full Disk Encryption (FDE) This is the strongest defense. It scrambles every bit on the drive so that without the key, the data is unreadable "garbage."
- "Software-Based: The OS handles encryption (e.g., BitLocker on Windows, FileVault on Mac, LUKS on Linux).
- Hardware-Based (SED): Self-Encrypting Drives have a dedicated chip that encrypts data as it's written. The CPU never sees the unencrypted data.

File System Permissions

- Once the drive is unlocked and running, the Operating System enforces rules using Access Control Lists (ACLs):
- Ownership: Every file belongs to a specific User or Group.
- Permissions: Defines who can Read, Write, or Execute a file.
- Root/Admin: Only the highest privilege level can bypass these rules to modify system files.

Hardware-Level Security

- TPM (Trusted Platform Module): A physical chip on your motherboard that stores the "Master Key." It ensures the drive only unlocks if the hardware hasn't been tampered with.
- ATA Passwords: A low-level password stored in the drive's firmware. If set, the drive will refuse to communicate with *any* computer until the password is provided.
- Write-Protect Switches: Common on SD cards or specialized industrial drives; it physically breaks the circuit to prevent writing data.

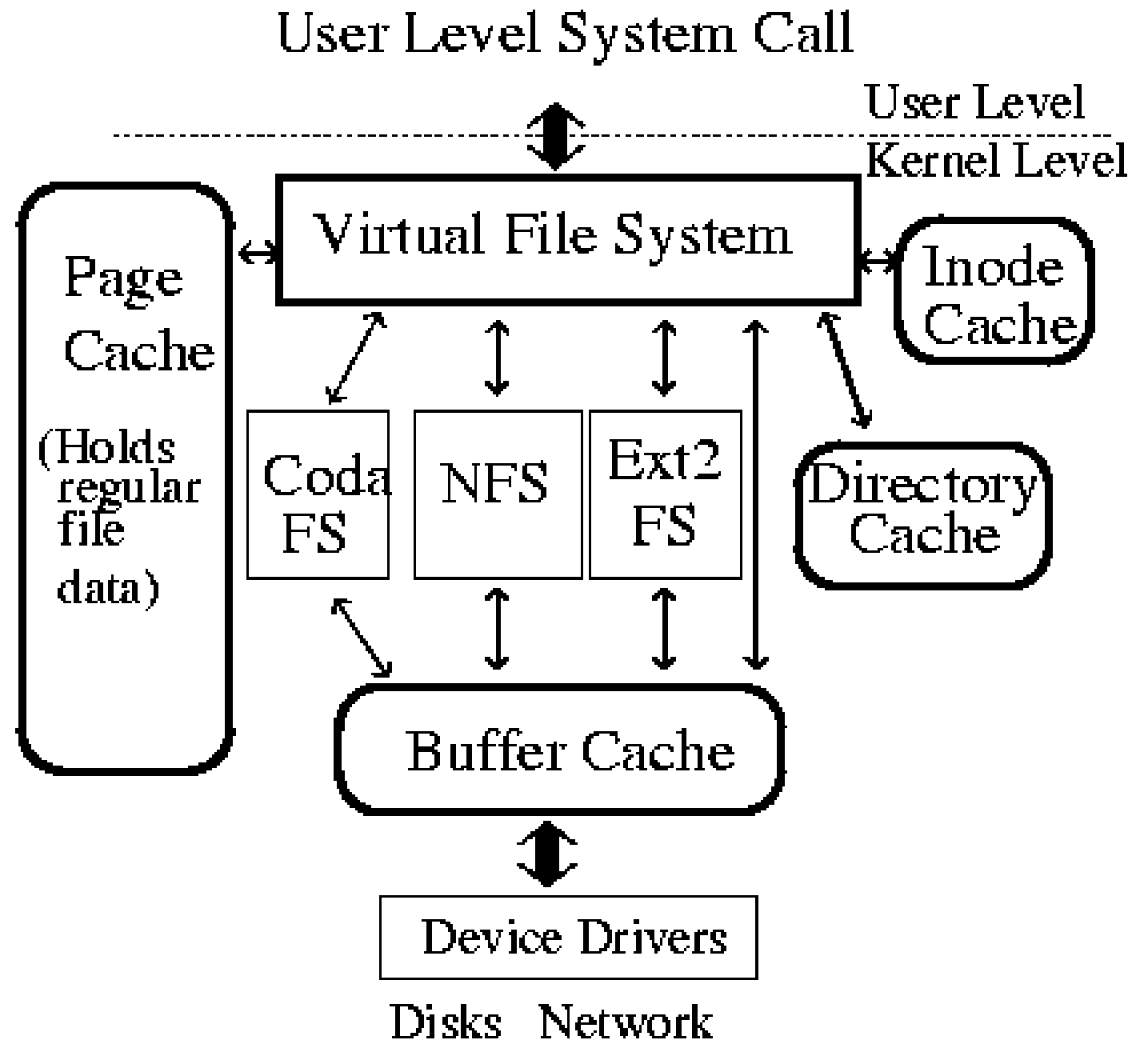
Logical Protection (Integrity)

- Checksums: The drive stores "math fingerprints" for data blocks. If bits flip (bit rot), the drive detects the mismatch.
- Journaling: File systems like NTFS or EXT4 keep a log of changes. If the power cuts, the log prevents the data from becoming "corrupted" or "orphaned."

Information protection in Linux

- Virtual File System (VFS) is a software layer within the Kernel that allows the OS to talk to different types of storage (hard drives, USB sticks, network drives) using the exact same commands
- Permission Bit Validation – The VFS is responsible for checking the standard Linux permission bits (rwxrwxrwx) stored in the Inode:
- When a process tries to `open()` a file, the VFS compares the process's Effective User ID (EUID) and Group ID (EGID) against the Inode's ownership bits.
- If the IDs don't match the required permissions, the VFS returns an `EACCES` (Permission Denied) error before the disk driver is ever contacted.

VFS interaction



Network security – The Kernel Firewall (Netfilter)

- Network connection: a pair of sockets $((IP_x, Port_x), (IP_y, Port_y))$
- In Linux, the core of network security is Netfilter, a framework inside the kernel that intercepts every packet.
- Tables & Chains: It uses rules (managed via nftables or iptables) to decide if a packet should be Accepted, Dropped, or Rejected.
- Stateful Inspection: The kernel tracks the "state" of connections. It knows if an incoming packet is a response to a request you sent, or an unsolicited (and potentially malicious) attempt to enter.

Encryption for Data in Transit

- To protect information from being "sniffed" on the wire, data is encrypted using cryptographic protocols:
- TLS (Transport Layer Security): Encrypts web and email traffic.
- IPsec: Provides security at the IP layer, often used for VPNs.
- SSH (Secure Shell): Encrypted management access to the OS.
- Hardware Acceleration: Modern x86_64 CPUs use the AES-NI instruction set to perform this encryption at the hardware level, ensuring security doesn't slow down the system.

Access Control & Authentication

- Security is only as good as the identity of the person accessing it:
- RBAC (Role-Based Access Control): Limiting users to only the network resources they need.
- MFA (Multi-Factor Authentication): Requiring a code or token in addition to a password.
- PAM (Pluggable Authentication Modules): A Linux mechanism that allows the system to use various authentication methods (biometrics, tokens, passwords) for network logins.

Some useful info

- Windows security is like a screen door on a submarine. It looks like it's doing something, but once the pressure is on, everyone is getting wet.
- Linux security is like a medieval castle. It's incredibly secure, but you have to build the drawbridge yourself, and if you lose the key, even the king can't get back in.
- MacOS security is like a high-end gated community. It's very safe, but the guards won't let you bring your own furniture inside because it doesn't match the aesthetic.